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Linear Statistical Models
BSMA3012

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(Q1)

Given Linear Model:

$$y_{ij} = \mu + \mu_i + \epsilon_{ij}$$

$$1 \leq i \leq 2 \ \& \ 1 \leq j \leq 2$$

Decide if μ & μ_i are identifiable from Model?

<u>Solution</u>

Number of parameters to be estimated $= 3 \ (\mu, \mu_1 \ \& \ \mu_2)$

Now

$$E(y_{1j}) = \mu + \mu_1$$

&

$$E(y_{2j}) = \mu + \mu_2$$

if we write the model in:

$$Y = X\beta + e$$

$$y_{11} = 1\cdot\mu + 1\cdot\mu_1 + 0\,\mu_2 + \epsilon_{11}$$

$$\vdots$$

$$y_{22} = 1\cdot\mu + 0\cdot\mu_2 + 1\cdot\mu_2 + \epsilon_{22}$$

$$Y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \end{pmatrix} \;\&\; X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \mu \\ \mu_1 \\ \mu_2 \end{pmatrix}, \quad \epsilon = \begin{pmatrix} \epsilon_{11} \\ \epsilon_{12} \\ \epsilon_{21} \\ \epsilon_{22} \end{pmatrix}$$

Now Rank of $X = 2$ $(C_1 = C_2 + C_3)$

as we see rank $= 2$
which is $<$ no. of parameters
to be estimated (3)

Therefore we cannot uniquely
identify μ, μ_1 & μ_2
together.

(Q2)

(a) $\underset{\sim}{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$; $\underset{\sim}{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}$

$\underset{\sim}{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \end{pmatrix}$ & $X = \begin{pmatrix} 1 & +2 & 3 & 4 \\ 1 & -2 & 3 & -4 \\ 1 & +2 & -3 & -4 \\ 1 & -2 & -3 & 4 \end{pmatrix}$

(b) for a linear model $(\underset{\sim}{Y}, X\underset{\sim}{\beta}, \sigma^2 I)$

$\hat{\beta}$ (estimate of β) can be given by:

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T Y}$$

$$\left[\begin{array}{c} Y = X\beta \\ \Rightarrow \boxed{X^T Y = (X^T X)\beta} \end{array} \right]$$

Now,

$$X^T X = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & -2 & 3 & -4 \\ 1 & 2 & -3 & -4 \\ 1 & -2 & -3 & 4 \end{pmatrix}^T \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & -2 & 3 & -4 \\ 1 & 2 & -3 & -4 \\ 1 & -2 & -3 & 4 \end{pmatrix}$$

$$=\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & -2 & 2 & -2 \\ 3 & 3 & -3 & -3 \\ 4 & -4 & -4 & 4 \end{pmatrix}\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & -2 & 3 & -4 \\ 1 & 2 & -3 & -4 \\ 1 & -2 & -3 & 4 \end{pmatrix}$$

$$=\begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 36 & 0 \\ 0 & 0 & 0 & 64 \end{pmatrix}$$

$$X^T y=\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & -2 & 3 & -4 \\ 1 & 2 & -3 & -4 \\ 1 & -2 & -3 & 4 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

$$=\begin{pmatrix} 1(y_1+y_2+y_3+y_4) \\ 2(y_1-y_2+y_3-y_4) \\ 3(y_1+y_2-y_3-y_4) \\ 4(y_1-y_2-y_3+y_4) \end{pmatrix}$$

So,

$$\begin{pmatrix} y_1+y_2+y_3+y_4 \\ 2(y_1-y_2+y_3-y_4) \\ 3(y_1+y_2-y_3-y_4) \\ 4(y_1-y_2-y_3+y_4) \end{pmatrix}=\begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 16 & 0 & 0 \\ 0 & 0 & 36 & 0 \\ 0 & 0 & 0 & 64 \end{pmatrix}\begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{pmatrix}$$

$$=\begin{pmatrix} 4\beta_1 \\ 16\beta_2 \\ 36\beta_3 \\ 64\beta_4 \end{pmatrix}$$

(c)

Using Normal eqⁿ (from b)

$$\hat{\beta}_1 = \frac{1}{4}(y_1 + y_2 + y_3 + y_4)$$

$$\hat{\beta}_2 = \frac{1}{16}(y_1 - y_2 + y_3 - y_4)$$

$$\hat{\beta}_3 = \frac{1}{36}(y_1 + y_2 - y_3 - y_4)$$

$$\hat{\beta}_4 = \frac{1}{64}(y_1 - y_2 - y_3 + y_4)$$

(Q3)

(a) Given,

$$y_1 = 15\beta_1 + 3\beta_2 + \epsilon_1$$
$$y_2 = 10\beta_1 - 2\beta_2 + \epsilon_2$$

So,

$$\underset{\sim}{Y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad X = \begin{bmatrix} 15 & 3 \\ 10 & -2 \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

$$\underset{\sim}{Y} - X\underset{\sim}{\beta} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} - \begin{bmatrix} 15 & 3 \\ 10 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} y_1 - 15\beta_1 - 3\beta_2 \\ y_2 - 10\beta_1 + 2\beta_2 \end{bmatrix}$$

as given

$$\|z\|_2^2 = z \cdot z = z^T z$$

now,

$$\|\underset{\sim}{y} - X\underset{\sim}{\beta}\|_2^2 = (y_1 - 15\beta_1 - 3\beta_2)^2 + (y_2 - 10\beta_1 + 2\beta_2)^2$$

$$\Rightarrow \|\underset{\sim}{y} - X\underset{\sim}{\beta}\|_2 = \sqrt{(y_1 - 15\beta_1 - 3\beta_2)^2 \quad (y_2 - 10\beta_1 + 2\beta_2)^2}$$

(b) for $\underset{\sim}{y} = \begin{pmatrix} 52 \\ 22 \end{pmatrix}$

$$\|\underset{\sim}{y} - X\underset{\sim}{\beta}\|_2 = \sqrt{(52 - 15\beta_1 - 3\beta_2)^2 + (22 - 10\beta_1 + 2\beta_2)^2}$$

(i) $\underset{\sim}{\beta} = (0, 1)^T$

$$\Rightarrow \|\underset{\sim}{y} - X\underset{\sim}{\beta}\|_2 = \sqrt{(52 - 0 - 3)^2 + (22 - 0 + 2)^2}$$

$$= \sqrt{49^2 + 24^2} = \sqrt{2977}$$

$$= 54$$

(ii) $\underset{\sim}{\beta} = (1, 0)^T$

$$\Rightarrow \|\underset{\sim}{Y} - X\underset{\sim}{\beta}\|_2 = \sqrt{(52-15)^2 + (22-10)^2}$$

$$= \sqrt{37^2 + 12^2} = \sqrt{1513}$$

$$= 38.89$$

(iii) $\underset{\sim}{\beta} = (1, 1)^T$

$$\|\underset{\sim}{Y} - X\underset{\sim}{\beta}\|^2 = \sqrt{(52-15-3)^2 + (22-10+2)^2}$$

$$= \sqrt{(52-18)^2 + (22-8)^2}$$

$$= \sqrt{34^2 + 14^2} = 34.2052$$

(iv) $\underset{\sim}{\beta} = (0.5, 0.5)^T$

$$\|\underset{\sim}{Y} - X\underset{\sim}{\beta}\|^2 = \sqrt{(52-7.5-1.5)^2 + (22-5+1)^2}$$

$$= \sqrt{43^2 + 18^2} = 46.6154$$

Since,

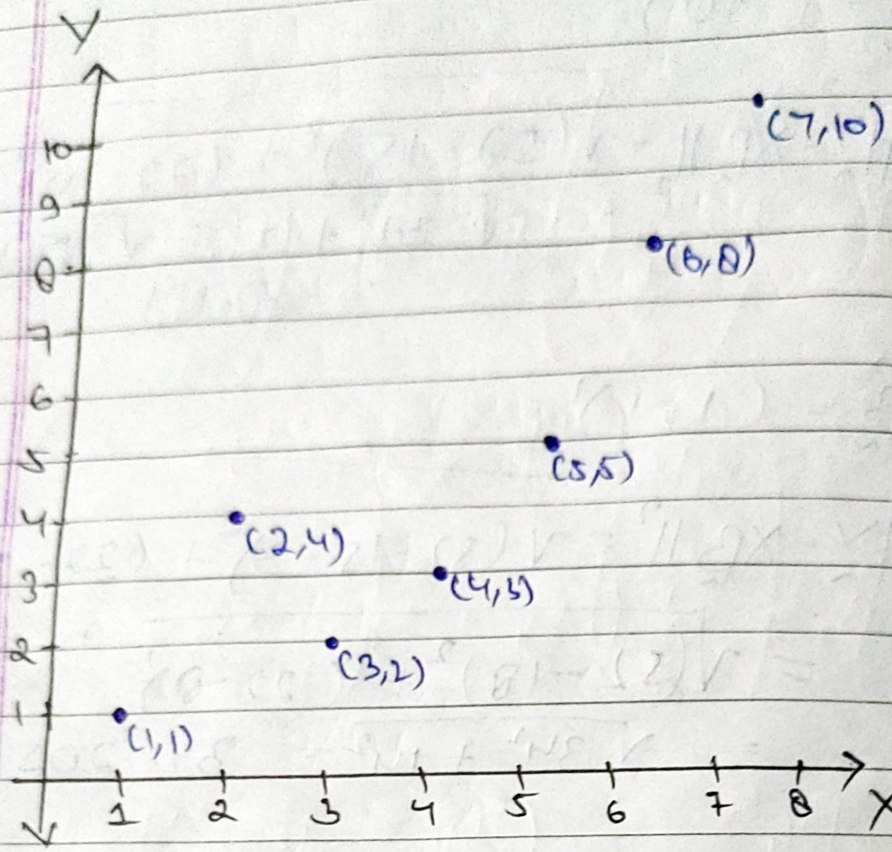
$$\|\underset{\sim}{Y} - X\underset{\sim}{\beta}\|_2 = 34.2052 \text{ (Minimum)}$$

$$(\beta = (1, 1)^T)$$

Therefore $\boxed{\text{(iii) } \beta = \binom{1}{1}}$ is correct

(4)

(a)



Scatter plot for given Dataset

(b) Least Sq. Estimate of $\underset{\sim}{\beta}$

We know that, using Normal Eqⁿ,

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

[Model ⇒ $Y = \beta_0 + \beta_1 X + \epsilon_i$]

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 4 \\ 2 \\ 3 \\ 5 \\ 8 \\ 10 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} n \times 1 & 1+2+\cdots+7 \\ 1+2+\cdots+7 & 1^2+2^2+\cdots+7^2 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 7 & 28 \\ 28 & 140 \end{bmatrix}$$

$$X^T Y = \begin{pmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{pmatrix} = \begin{pmatrix} 33 \\ 170 \end{pmatrix}$$

$$(X^T X)^{-1} = \frac{1}{\det|X^T X|} \operatorname{adj}(X^T X)$$

$$= \frac{1}{28^2 - 7 \cdot 140} \begin{bmatrix} 140 & -28 \\ -28 & 7 \end{bmatrix}$$

$$= \frac{1}{196} \begin{bmatrix} 140 & -28 \\ -28 & 7 \end{bmatrix}$$

$$\hat{\beta} = \frac{1}{196}\begin{pmatrix} 140 & -28 \\ -28 & 7 \end{pmatrix}\begin{pmatrix} 33 \\ 170 \end{pmatrix}$$

$$= \frac{1}{196}\begin{pmatrix} 4620 - 4760 \\ -924 + 1190 \end{pmatrix}$$

$$= \frac{1}{196}\begin{pmatrix} -140 \\ 266 \end{pmatrix}$$

$$\Rightarrow \hat{\beta}_0 = -5/7$$

$$\Rightarrow \hat{\beta}_1 = 19/14$$

$$\hat{\beta} = \begin{pmatrix} -5/7 \\ +19/14 \end{pmatrix}$$

(Q5) $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$

$$RREF(A) = \begin{pmatrix} 1 & 2 \\ 3-3\cdot 1 & 6-3\cdot 2 \end{pmatrix}$$

(by $R_2 \longrightarrow R_2 - 3R_1$)

Pivot $\begin{pmatrix} \boxed{1} & 2 \\ 0 & 0 \end{pmatrix}$

As we know,

Row Space (A) = Row Space of RREF(A)

$\Rightarrow R(A)$ or $C(A^T) = \text{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$

<u>Colum space of A.</u>

Colum space = span of columns from original matrix which have pivot element

here C_1 has pivot therefore

$C(A) = \text{span}\left\{\begin{pmatrix}1\\3\end{pmatrix}\right\}$

<u>Null space (A):</u>

$$\begin{pmatrix}1 & 2\\0 & 0\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix} = 0$$

$\Rightarrow 2x_2 + x_1 = 0$

$\Rightarrow x_1 = -2x_2$

Null space = $\text{span}\left\{\begin{matrix}-2\\1\end{matrix}\right\}$ for any x_2

Left Null Space:

$$\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$v_1 + 3v_2 = 0$
$2v_1 + 6v_2 = 0$

$v_1 = -3v_2$

assume, $v_2 = \lambda$
then
$v_1 = -3\lambda$

soln $= \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} -3\lambda \\ \lambda \end{pmatrix} = \lambda\begin{pmatrix} -3 \\ 1 \end{pmatrix}$

Null Space(A^T) = Span$\left\{\begin{pmatrix} -3 \\ 1 \end{pmatrix}\right\}$